

QoL - Data imputation

Setup

```
library(here)
library(tidyverse)
library(patchwork)
library(mice)
library(miceadds)
library(arrow)
```

Load data

These are historic data from patients of the Division of Medical Oncology at USB, which had their first appointment from 2023 to mid 2025, i.e., before LIQPLAT started. The data are not publicly available.

```
set.seed(1234)
df <- read_parquet(here("data", "alt_qol.parquet"))
```

Simplify some baseline variables

```
# remove observations after day 182
# lump factor variables according to frequency to have a maximum of 5 levels

df <- df |>
  group_by(pat_id) |>
  filter(quest_day <= 182) |>
  ungroup() |>
  mutate(
    diagnosis = fct_lump_n(diagnosis, n = 4),
    diagnosis = factor(as.integer(diagnosis)),
    plan_fstcnt_coded = fct_lump_n(plan_fstcnt_coded, n = 4),
    plan_fstcnt_coded = factor(as.integer(plan_fstcnt_coded)),
    status      = if_else(!is.na(death_day) & death_day <= 182, 1, 0),
    time        = if_else(!is.na(death_day) & death_day <= 182, death_day, 182),
    q30         = factor(q30, levels = 1:7, ordered = TRUE),
    ecog_fstcnt = factor(ecog_fstcnt, levels = 1:4, ordered = TRUE),
  ) |>
  select(pat_id, pat_age, gender, ecog_fstcnt, diagnosis, plan_fstcnt_coded, time,
    status, quest_day, q30) |>
  arrange(pat_id, quest_day)
```

Variables for final analysis

When analyzing the trial data, we will use the following predictors for imputation and modeling:

Demographics and baseline clinical:

- tx - treatment group allocation
- pat_age - age at enrollment

- gender - patient sex
- ecog_fstcnt - ECOG performance status (ordered factor: 0-4)
- diagnosis - primary cancer diagnosis (lumped to 5 most frequent categories)

Laboratory values:

- albumin - serum albumin (g/L)
- c_reactive_protein - CRP (mg/L)

Healthcare utilization:

- n_stay_plus_ER - total days in hospital or emergency room
- n_blood_products_all - number of blood product units received

Clinical events:

- event_next_treatment - indicator for starting next line of treatment
- event_progression - indicator for disease progression

Outcome:

- status - death indicator
- na_est - Nelson-Aalen cumulative hazard estimate

Longitudinal outcome:

- quest_day - day of questionnaire completion
- q30 - EORTC QLQ-C30 overall quality of life (ordered factor: 1-7)

The laboratory values and healthcare utilization variables are currently unavailable in the dummy dataset.

Add Nelson-Aalen-estimates, Randomize, and add missing on day 0

We need to consider the day of random selection as day 0, otherwise we immortalize patients. This creates a new problem: people who die before the day of their first contact, even though they were randomly selected, would be deleted from the analysis dataset because of a missing yprev before their death observation. Because we need to get an accurate cumulative incidence for death, we need to impute a plausible QoL value for these patients before their death, e.g., at the day of random selection. We also need a value for day 0 for the marginalization. MI is probably better than carrying the first observed value backwards.

I expect that 50% of patients are randomized on their date of first contact, and 50% before, but that death before the first contact will be exceedingly rare.

```
# nelson aalen estimator
df_na_est <- df |> select(pat_id, time, status) |> distinct()

na_est <- nelsonaalen(df_na_est, timevar = time, statusvar = status)
na_est <- tibble(pat_id = df_na_est$pat_id, na_est)

df <- df |> left_join(na_est, by = "pat_id")

# create random group allocation.
tx <- sample(unique(df$pat_id), size = 0.5 * length(unique(df$pat_id)), replace = FALSE)
```

```

df <- df |>
  mutate(tx = if_else(pat_id %in% tx, "1", "0"))

id_cols <- c(
  "pat_id", "pat_age", "gender", "ecog_fstcnt", "diagnosis",
  "plan_fstcnt_coded", "time", "status", "na_est", "tx"
)

# ensure that everyone has a questionnaire at baseline
empty_obs <- df |>
  group_by(pick(all_of(id_cols))) |>
  complete(quest_day = union(quest_day, 0)) |>
  ungroup() |>
  mutate(week = ceiling(quest_day/7))

```

Multiple imputation

Baseline values will be imputed at the patient (cluster level). This ensures that only one value per multiple rows of missing data for a single patient is imputed. We also impute missing responses to q30.

```
md.pattern(empty_obs)
```

```

init <- mice(empty_obs, maxit = 0)
pred_matrix <- init$predictorMatrix
meth <- init$method

# don't want to use survival time because we use na_est.
pred_matrix[, "time"] <- 0 # 'time' will not predict any other variable
pred_matrix["time", ] <- 0 # 'time' will not be predicted by any other variable
meth["time"] <- ""

# Use patient ID as clustering variable
pred_matrix[, "pat_id"] <- -2

# Impute individual q30 at the questionnaire level
meth["q30"] <- "2l.pmm"

# use mean matching at the cluster level for baseline observations
# this ensures only one ecog per patient per imputation
meth["ecog_fstcnt"] <- "2lonly.pmm"
meth["plan_fstcnt_coded"] <- "2lonly.pmm"
meth["diagnosis"] <- "2lonly.pmm"

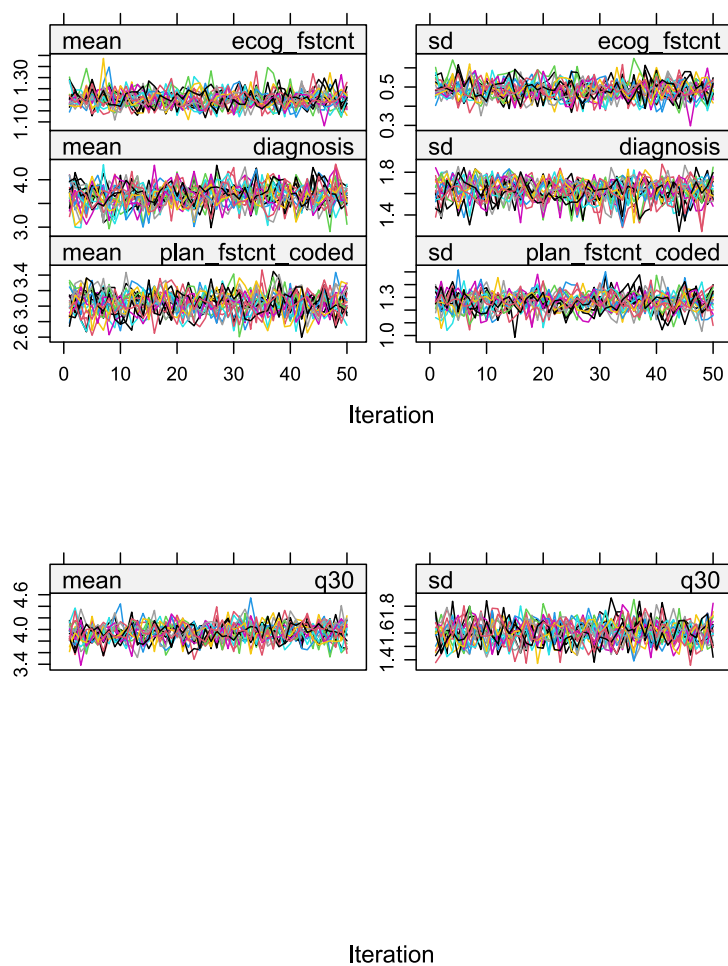
imputed_data <- mice(empty_obs,
  m = 20, # 50 in final analysis
  maxit = 50,
  predictorMatrix = pred_matrix,
  method = meth,
  seed = 78109823)

```

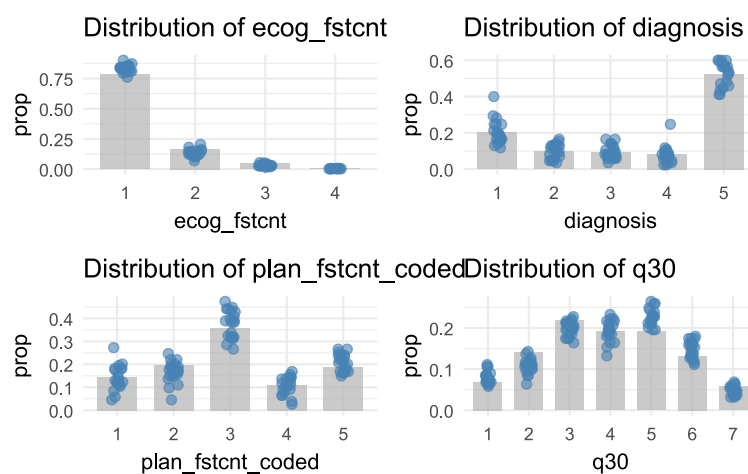
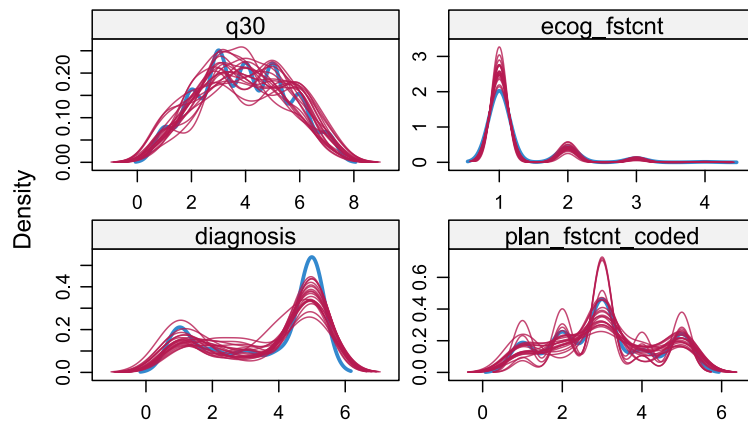
Diagnostics

The diagnostics look fine to me.

```
plot(imputed_data)
```



```
densityplot(imputed_data, ~q30 + ecog_fstcnt + diagnosis + plan_fstcnt_coded)
```



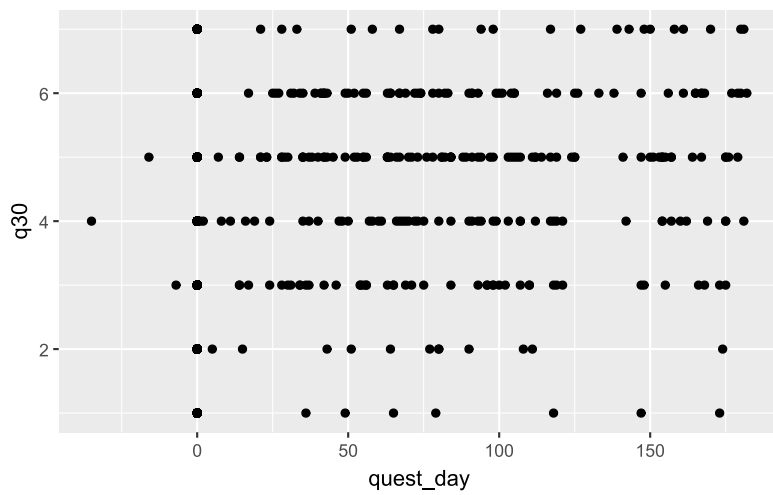
Select one imputation

For the analysis of the trial data we will use all the imputed datasets and stack the posterior draws. However, for our simulations this is unfeasible, and we shall thus only use one complete dataset for the setup.

We can see that we have data for most days of follow-up, with a bit of scarcity from day 125 to day 150.

```
complete_data <- complete(imputed_data, 4)

complete_data |>
  mutate(q30 = as.numeric(q30)) |>
  ggplot(aes(x = quest_day, y = q30)) +
  geom_point()
```



Export

```
write_parquet(complete_data, sink = here("data", "qol_complete.parquet"))
```